Geospatial analysis of Wildfires in Nevada

Muhammad Soban Ijaz (BSCS 22064)

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Supervisor: Dr. Adnan Siddique

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1. Introduction

Wildfires are a growing environmental threat, particularly in arid regions like Nevada, where climatic factors such as high temperature and low precipitation contribute significantly to fire risk. This project aims to explore the spatial and temporal relationship between wildfire occurrences and environmental variables (temoerature, precipitation i.e which will give drought trends and vapor pressure deficit i.e gives information of air dryness)—from 2020 to 2022.

2. Problem Statement

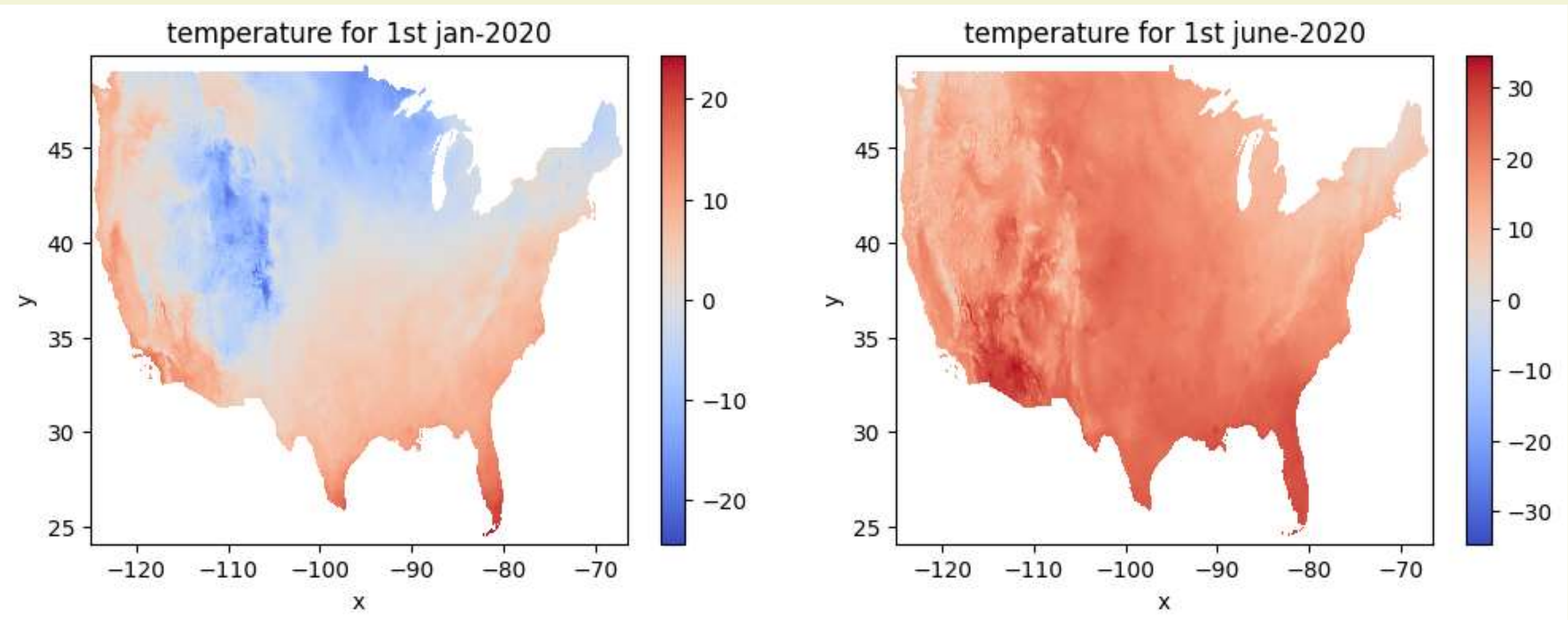
The project aims to find the influence of environmental variables-temperature, precipitation and vapor pressure deficit-with wildfires.

- Are wildfires directly or inverly proportional to environmental variables? Do wildfires tend to cluster together or not?

3. Data

Nevada wildfire occurrence data which includes wlfires occurred in Nevada from 2013 to 2024.Rastor data-Temperature,precipitation and vapor pressure deficit. Daily Temperature data for 2020-2022. Monthly precipitation mean values (2020-2022) and monthly vapor pressure deficit values (2020-2022).

- Preprocessing:
- Wildfires data was queried for 2020-2022 and split on the bases of years with extra columns removed.For each of these the zonal stats were computed to include temperature,precipitation and vapor pressure deficit records on each location fire occurred.Rastor data was dropped of Nan values.



4. Methodology

Spatial lag was employed to understand the influence of nearby locations To understand how these variables influence wildfires global spatial autocorrletation was employed , local spatial auto-correlation was employed to detect whether the wildfires tend to occur in Hotspots , Coldspots , doughnuts or diamonds in the rough. Ripley G function were plotted to find whether there exits some degree of clustering whithin wildfires and clustering techniques were used to identify clusters. We further enhance this analysis by incorporating spatial regression models to quantify how climate variables explain fire behavior, and use regionalization to classify Nevada into zones of similar wildfire–climate dynamics.

5. Results & Discussion

Spatial lag gave information on how nearby locations were affecting the observations giving spatial outliers and quantifing spatial auto-correlation i.e positive or negative . Global spatial auto-correlation indicated that the relation between temperature , vapor pressure deficit and precipitation with wildfires was positive i.e similar values close to other similar values. Local spatial auto-correlation suggested that for 2020 and 2021 wildfires occurred more in areas of high temperatures but in 2022 more wildfires were observed in low temperature areas which could indicate some other social factors effecting wildfires for that year. Wildfires tend to occurr in areas where there was very low rainfall and very high vapor pressure deficit indicating more dryness in air. Spatial regression suggested that there was some kind of spatial dependance in the data due to which simple linear regression did not work well , the clustering of residuals was discounted after incoporating spatial regression with spatial fixed effects and spatial lag model indicating wildfires vary with neighborhood and the nearby location has an influence on the current observation. Clustering was observed after applying Ripley G indicating location and environmental variables have an influence on the occurrence of wildfire. The work can be extended to find relation of more factors with wildfires like camping sites and bornfires.

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